

Assessment Task 1 (AT1)– Part 2

DevSecOps RiskDetective Challenge Scenario-Based Collaborative Analysis

TEAM – 5

TITLE: Airline Reservation System

Background:

Sky Fly Airlines continuously updates its online booking platform with promotional offers and new services. The application depends heavily on open-source libraries but no formal process exists for checking known vulnerabilities. Patches are delayed to avoid interrupting business operations, API protection is inconsistent and security reviews are reactive. A recent outage disrupted online ticket bookings, causing revenue loss and customer complaints.

TEAM MEMBERS

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Investigation Questions:

1. Identify at least five security risks.

Security Risk	Explanation
1. Unpatched Software	Delayed updates allow attackers to exploit known security flaws.
2. Vulnerable Open-Source Libraries	Third-party libraries may contain publicly known vulnerabilities.
3. Weak API Security	Inconsistent API protection can allow unauthorized users to access airline services
4. Service Outages (Downtime)	Cyberattacks or system failures can stop online ticket booking, causing revenue loss.
5. Data Breach	Customer information such as names, passport numbers, payment details, and contact information may be stolen.

2. Identify at least five vulnerabilities and classify each as High, Medium or Low.

Vulnerability	Description	Risk Level
Outdated Open-Source Libraries	Libraries contain known vulnerabilities.	high
Delayed Security Patches	Systems remain exposed to attacks.	high
Weak API Authentication	Attackers can misuse APIs to access sensitive functions.	high
No Continuous Security Monitoring	Threats are detected only after an attack.	medium
Poor Vulnerability Management Process	No regular scanning or prioritisation of vulnerabilities.	medium

3. List possible cyber threats and explain their impact.

Cyber Threat	Impact
SQL Injection	Attackers steal or modify airline database information.
Distributed Denial of Service (DDoS)	Website becomes unavailable, preventing ticket bookings.
API Attacks	Attackers manipulate booking or payment services.
Ransomware	Airline systems become encrypted and unavailable until a ransom is paid.
Phishing	Employee credentials are stolen, allowing unauthorised access.

4. DevOps vs DevSecOps (Minimum Eight Points)

DevOps

Focuses on fast software delivery.

Security is tested at the end.

Developers and operations collaborate.

Security testing is mostly manual.

Vulnerabilities are fixed later.

Less emphasis on compliance.

Security tools are optional.

Higher chance of vulnerabilities reaching production.

DevSecOps

Focuses on secure and fast software delivery.

Security is integrated throughout development.

Developers, operations, and security collaborate.

Security testing is automated.

Vulnerabilities are fixed immediately.

Compliance is built into the development process.

Security tools are mandatory in the pipeline.

Lower chance of vulnerabilities reaching production.

5. Recommend Five OWASP Security Best Practices

1. Regular Dependency Scanning

- Scan open-source libraries regularly.
- Replace vulnerable components immediately.

2. Strong Authentication

- Implement Multi-Factor Authentication (MFA).
- Use strong password policies.

3. Secure API Protection

- Use authentication tokens (JWT or OAuth).
- Apply rate limiting to prevent abuse.

4. Input Validation

- Validate all user inputs.
- Prevent SQL Injection and Cross-Site Scripting (XSS).

5. Continuous Security Testing

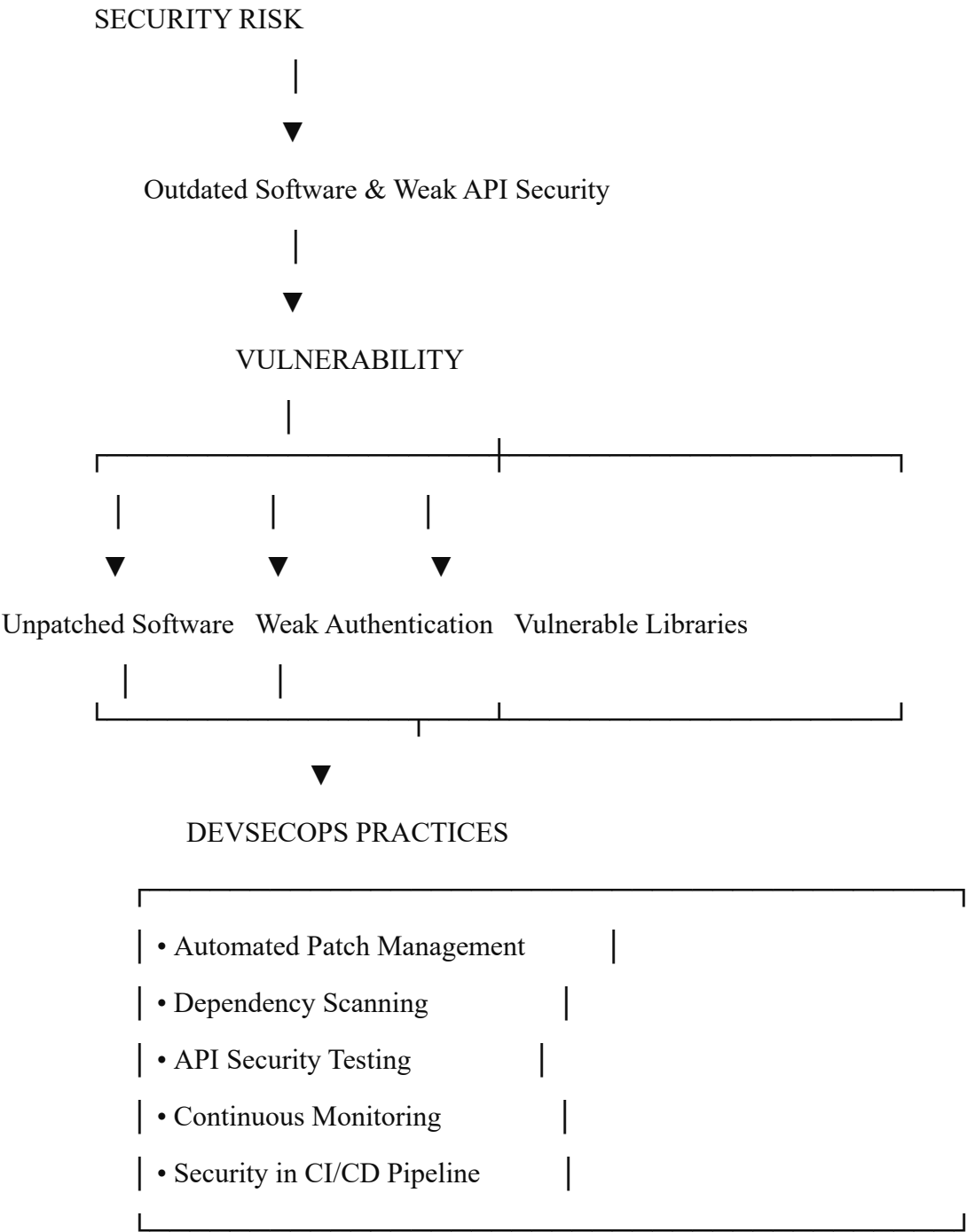
- Perform automated vulnerability scanning.
- Integrate security testing into the CI/CD pipeline.

6. Explain the Business Impact if These Issues Remain Unresolved

If these security issues are not resolved, the airline may experience serious business consequences:

- **Revenue Loss** due to booking system downtime.
- **Customer Dissatisfaction** because customers cannot book or manage flights.
- **Loss of Reputation** resulting from security incidents.
- **Legal and Regulatory Penalties** for failing to protect customer data.
- **Financial Loss** due to recovery costs, compensation, and fines.
- **Data Theft** involving customer personal and payment information.
- **Reduced Customer Trust**, leading to fewer future bookings.

7.conceptmap





BUSINESS BENEFITS

• Better Security	
• Reduced Downtime	
• Increased Customer Trust	
• Higher Revenue	
• Regulatory Compliance	
• Faster Incident Response	

8. Prepare a five-minute presentation summarizing your findings.

Slide 1 – Title

Airline Reservation System Security Analysis

Team 5

Slide 2 – Current Problems

- Heavy use of open-source libraries
- Delayed security patching
- Weak API security
- Reactive security reviews
- Recent outage affected ticket booking

Slide 3 – Security Risks & Vulnerabilities

Risks

- Unpatched software
- Weak APIs

- Data breaches
- Service downtime
- Vulnerable libraries

Vulnerabilities

- Outdated libraries (High)
- Delayed patches (High)
- Weak API authentication (High)
- Poor monitoring (Medium)
- Weak logging (Low)

Slide 4 – Cyber Threats

- SQL Injection
- DDoS Attacks
- API Abuse
- Ransomware
- Phishing
- Supply Chain Attacks
- Credential Stuffing

Impact:

- Revenue loss
- Data theft
- Customer dissatisfaction
- Reputation damage

Slide 5 – DevSecOps & OWASP Recommendations

DevSecOps

- Security integrated into CI/CD
- Automated vulnerability scanning
- Continuous monitoring

OWASP Best Practices

- Dependency scanning
- Strong authentication
- API security
- Input validation
- Continuous security testing

Slide 6 – Business Benefits

- Improved security
- Fewer cyberattacks
- Reduced downtime
- Better customer trust
- Regulatory compliance
- Increased revenue

Conclusion

Implementing **DevSecOps** and following **OWASP security best practices** enables SkyFly Airlines to identify vulnerabilities early, secure APIs and open-source components, minimise service outages, protect customer data, and maintain business continuity.

THANK YOU